Week 3:
Data Preprocessing and Regression Analysis

WIA1006 | Semester 2 | Tutorial Session | University of Malaya

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Questions and Solutions

Linear Regression (Univariate)

Question #1

Question #1:

1. Consider the problem of predicting sunny weather condition in each week of 2023 given the sunny weather condition in each week of 2022.
   In this scenario, x represents the number of days in each week that the weather is sunny in 2022. The value of y is defined as "the number of sunny days" in each week of 2023 which we want to predict.
   The following training set is a sample of few weeks with number of sunny days in each of them.
   Recall that in linear regression our hypothesis is $h_\theta(x) = \theta_0 + \theta_1 x$, and we use *m* to denote the number of training examples.

x	y
3	2
2	3
4	5
1	1
5	4

For the training set given above, what is the value of m?

Question

Linear Regression (Univariate)

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x	y
3	2
2	3
4	5
1	1
5	4

For the training set given above, what is the value of m?

Ans: 5

Key idea

m = number of training examples

Each row in the (x, y) table is one training example.

Training set table

x	y
3	2
2	3
4	5
1	1
5	4

5 rows = 5 training examples

Therefore, m = 5

Question #2

Question #2:

2. For this question, continue using the data provided in (1). Recall the definition of cost function for linear regression is

$$J(\theta_0, \theta_1) = \frac{1}{2m} * \sum_{i}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2$$

What is $J(0,1)$?

Linear Regression (Univariate) - Cost Function Example

Continue using the training data from Question 1. Compute the linear regression cost function J(0,1).

Cost function for linear regression

$$J(\theta_0, \theta_1) = 1/(2m) \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$$

Given: $\theta_0 = 0$ and $\theta_1 = 1$
So $h\theta(x) = \theta_0 + \theta_1 x = x$

Step-by-step calculation

x	y	hθ(x)=x	Error hθ(x)-y	Squared error	Value
3	2	3	1	1^2	1
2	3	2	-1	$(-1)^2$	1
4	5	4	-1	$(-1)^2$	1
1	1	1	0	0^2	0
5	4	5	1	1^2	1

Sum of squared errors = 1 + 1 + 1 + 0 + 1 = 4

Tip: each row is one training example, compute prediction, error, square the error, then average using 1/(2m).

Training data

x	y
3	2
2	3
4	5
1	1
5	4

m = 5 examples

Final computation

J(0,1) = 1/(2·5) · 4

= 4/10

= 0.4

Answer: J(0,1) = 0.4

Question #3

Question #3:

3. Suppose we set $\theta_0 = -1$, $\theta_1 = 0.5$, what is $h_\theta(3)$?

Linear Regression (Univariate)

Given θ_0 = -1 and θ_1 = 0.5, find hθ(3)

Use the hypothesis formula:

$$h_\theta(x) = \theta_0 + \theta_1 x$$

Substitute x = 3

$$h_\theta(3) = -1 + 0.5(3)$$
$$= -1 + 1.5$$
$$= 0.5$$

Key idea

- θ_0 is the intercept.
- θ_1 is the slope.
- Replace x with 3 and simplify step by step.
- Final answer: hθ(3) = 0.5

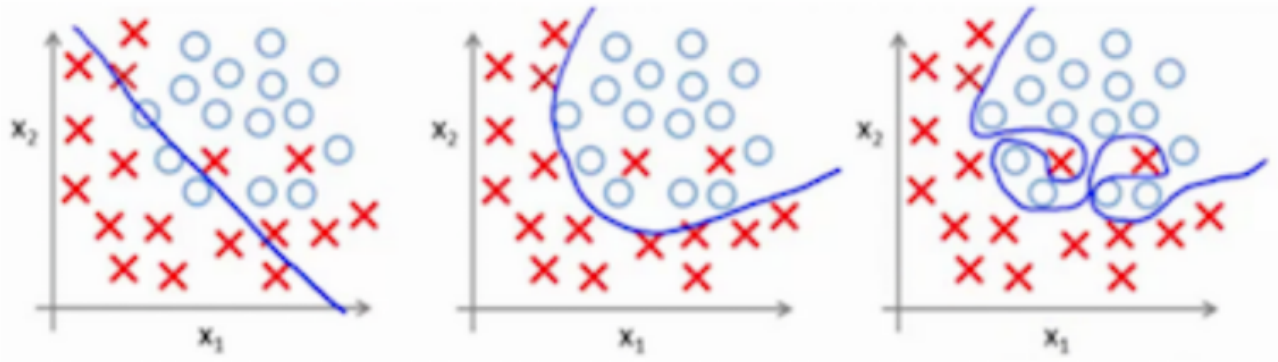
Answer: 0.5

Question #4

Question #4:

4. Three different classifiers are trained on the same data. Their decision boundaries are shown below. Which of the following statements are true?

UNDERFITTING BEST FIT OVERFITTING



✔ The leftmost classifier has high robustness, poor fit.

€ The leftmost classifier has poor robustness, high fit.

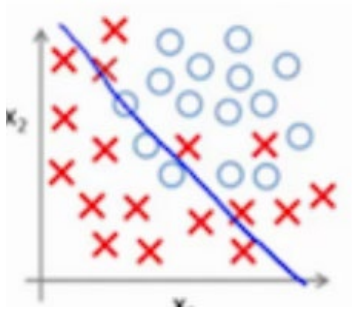
✔ The rightmost classifier has poor robustness, high fit.

€ The rightmost classifier has high robustness, poor fit

Q4: Underfitting vs Best Fit vs Overfitting

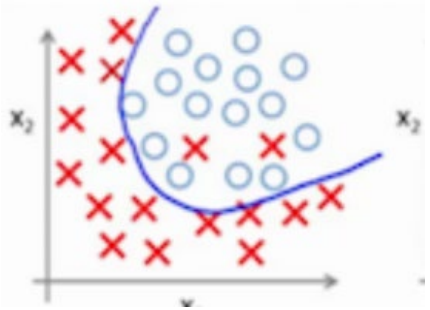
Which statements are true about robustness and fit?

UNDERFITTING



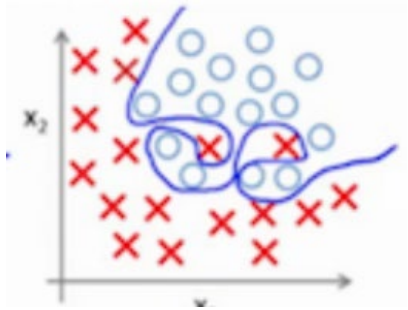
Too simple → high robustness, but poor fit.

BEST FIT



Balances pattern learning and generalization.

OVERFITTING



Fits training data too closely → poor robustness.

Correct takeaways

- ✓ The leftmost classifier has high robustness, poor fit.
- ✓ The rightmost classifier has poor robustness, high fit.

Tip: underfitting = too simple, overfitting = too complex.

Question #5

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Question #5:

5. What is the difference between local minima and global minima gradient descent?

https://blog.paperspace.com/intro-to-optimization-in-deep-learning-gradient-descent/#:~:text=Local%20minimum%20are%20called%20so,entire%20domain%20the%20loss%20function

Linear Regression (Multivariate)

Question #1

Question #1:

1. Suppose we have m = 4 houses, and each house has area and number of bedrooms which can be used to predict the house price. A dataset of the features is as follows:

	Bedrooms	Sqft_area	Price
1.	1	880	490,000
2.	3	1930	630,000
3.	4	1940	640,000
4.	3	1350	570,000

You'd like to use polynomial regression to predict a house price from its numbers of bedrooms and sqft_area. Concretely, suppose you wish to fit a model of the form.

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

where x_1 is the number of bedrooms and x_2 is sqft_area. Further you plan to use both feature scaling (dividing by the max-min, or range, of a feature) and mean normalization.

What is the normalized feature x_2^4(i.e. for the fourth training data)?

Feature Scaling + Mean Normalization Example

Find the normalized feature $x_2^{(4)}$ for the 4th house, where x_2 is Sqft_area.

Dataset

No.	Bedrooms	Sqft_area	Price
1	1	880	490,000
2	3	1930	630,000
3	4	1940	640,000
4	**3**	**1350**	**570,000**

Highlighted row = 4th house

Sqft_area values

880, 1930, 1940, 1350

Mean = (880 + 1930 + 1940 + 1350) / 4 = 1525

Range = max - min = 1940 - 880 = 1060

Normalization formula

x_norm = (x - μ) / (max - min)

For the 4th house: $x_2^{(4)} = 1350$

Step-by-step substitution

$$x_2^{(4)} = (1350 - 1525) / (1940 - 880)$$

$$= -175 / 1060$$

$$\approx -0.165 \quad \text{(about } -0.16\text{)}$$

Answer: the normalized Sqft_area for the 4th house is approximately -0.16.

Question #2

Question #2:

2. You run gradient descent for 12 iterations with $\alpha = 0.2$ and compute $J(\theta)$ after each iteration you find that the value of $J(\theta)$ increases over time. What would you do to correct this issue?

   Ans:
   Since if α is too large $J(\theta)$ may not decrease on every iteration and thus may not converge.
   It'd be more promising to try a smaller value of α (e.g., $\alpha = 0.1$).

Tip: Increasing cost usually means the learning rate is too big, so gradient descent fails to converge.

Question #3

Question #3:

3. Suppose you have m = 23 training examples with n = 5 features (excluding the additional all-ones feature for the intercept term, which you should add). The normal equation is $\theta=(X^TX)^{-1}X^T y$. For the given values of m and n, what are the dimensions of θ, X, and y in this equation?

   1. X is 23 × 6, y is 23 × 6, θ is 6 × 6
   2. X is 23 × 5, y is 23 × 1, θ is 5 × 1
   3. X is 23 × 6, y is 23 × 1, θ is 6 × 1
   4. X is 23 × 6, y is 23 × 1, θ is 5 × 5

Normal Equation: Dimensions of X, y, and θ

Given: m = 23 training examples, n = 5 features, plus 1 intercept column

Step 1: Add the intercept term

Total columns in X = n + 1 = 5 + 1 = 6

X	θ	y
Design matrix	Parameters	Targets
23 × 6	**6 × 1**	**23 × 1**

Step 2: Check the normal equation

$$\theta = (X^T X)^{-1} X^T y$$

This only works when dimensions are consistent:

X: 23×6, y: 23×1, θ: 6×1

Correct option: 3

Answer: X is 23×6, y is 23×1, θ is 6×1

Question #4

Question #4:

4. Suppose you have a dataset with m = 1000000 examples and n = 200000 features for each example. You want to use multivariate linear regression to fit the parameters θ to our data. Should you prefer gradient descent or the normal equation?

   Ans: Gradient descent, since $(X^TX)^{-1}$ will be very slow to compute in the normal equation. With n = 200000 features, you have to invert a 200001 x 200001 matrix to compute the normal equation. Inverting such a large matrix is computationally expensive, so gradient descent is a good choice.

Question #5

Question #5:

5. Which of the following statements are true?
   1. MAE doesn't add any additional weight to the distance between points. The error growth is linear.
   2. MSE errors grow exponentially with larger values of distance. It's a metric that adds a massive penalty to points that are far away and a minimal penalty for points that are close to the expected result.
   3. It is necessary to prevent the normal equation from getting stuck in local optima.
   4. It prevents the matrix XTX (used in the normal equation) from being non-invertable (singular/degenerate).

   Ans: 1 and 2

Record your attendance

Thank You!

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